

KL Divergence and Maximum Likelihood

$$D_{KL}(P, Q) = \sum_{x \in X} p(x) \log \left[\frac{p(x)}{q(x)} \right]$$

1. prove $D_{KL}(P, Q) \geq 0$

$$D_{KL}(P, Q) = \sum p(x) \log \left[\frac{p(x)}{q(x)} \right]$$

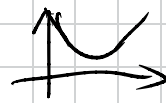
$$= \sum p(x) (\log p(x) - \log q(x))$$

$$= \sum p(x) \log p(x) - \sum p(x) \log q(x)$$

$$= -\sum p(x) \log q(x) + \sum p(x) \log p(x)$$

$$= -\sum p(x) \log \left[\frac{q(x)}{p(x)} \right] = \underbrace{E_{p(x)} \left[-\log \left(\frac{q(x)}{p(x)} \right) \right]}_{\Delta}$$

Jensen's Inequality



for convex : $E[f(x)] \geq f(E[X])$

$$\Rightarrow E[-\log(x)] \geq -\log(\underbrace{E[X]})$$

$$\Rightarrow D_{KL}(P, Q) \geq -\log(E_{P(X)}[\frac{q(x)}{p(x)}])$$

$$= -\log(\sum_i p_i \cdot \frac{q_i}{p_i})$$

$$= -\log(\sum_i q_i)$$

$$= -\log(1)$$

$$= 0$$

$$\Rightarrow D_{KL}(P, Q) \geq 0$$

2. Prove $D_{KL}(P, Q) = 0$ iff $P(x) = Q(x)$

2.1 If $D_{KL}(P, Q) = 0$, then $P(x) = Q(x)$

Jensen's : f is strictly convex, $f(x) = -\log(x)$

Z is random var,

$$\text{and } E[f(Z)] \geq f(E[Z])$$

then $Z = E(Z)$, Z is a non random constant.

From Part 1 : $D_{KL}(P, Q) \geq -\log(1) = 0$

if $D_{KL} = 0$

$$\text{then : } E_P[-\log(\frac{Q}{P})] = -\log(E_P[\frac{Q}{P}])$$

$$\Rightarrow E[f(Z)] = -\log(E[Z]), \quad Z = \frac{Q(x)}{P(x)}$$

$$\Rightarrow E[f(Z)] = f(E[Z])$$

Due to strict convex, Z must be constant here.

$$\text{so } \frac{Q(x)}{P(x)} = c \text{ for all } x.$$

$$\text{since } \mathbb{E}_p \left[\frac{q}{p} \right] = \sum p(x) \cdot \frac{q(x)}{p(x)} = \sum q(x) = 1$$

$$\text{with } \frac{q(x)}{p(x)} = c, \quad \sum p(x) \cdot c = c \cdot 1 = c$$

$$\text{since } \frac{q(x)}{p(x)} = 1, \quad q(x) = p(x) \checkmark$$

$$2.2 \text{ if } p(x) = q(x), \quad D_{KL} = 0$$

$$D_{KL}(p, q) = \sum p(x) \log \left[\frac{p(x)}{q(x)} \right]$$

$$\text{if } p(x) = q(x), \text{ then } \frac{p(x)}{q(x)} = 1$$

$$\Rightarrow \log \left[\frac{p(x)}{q(x)} \right] = \log(1) = 0$$

$$\Rightarrow D_{KL}(p, q) = \sum p(x) \cdot \underline{0} = 0$$

3. Prove $\arg \min_{\theta} D_{KL}(P_{data}, P_{\theta}) = \arg \max_{\theta} \sum_{i=1}^n \log(P_{\theta}(\tilde{x}^i))$

$$D_{KL}(P_{data}, P_{\theta}) = \sum_x P_{data}(x) \log \left[\frac{P_{data}(x)}{P_{\theta}(x)} \right]$$

$$= \sum_x P_{data}(x) \cdot \log P_{data}(x) - \sum_x P_{data}(x) \log P_{\theta}(x)$$

$$= \text{Entropy}(P_{data}) - \sum_x P_{data}(x) \log P_{\theta}(x)$$

$$\sum_x P_{data}(x) \log P_{\theta}(x) = \sum_{i=1}^n P_{data}(x^i) \log P_{\theta}(x^i)$$

$$\text{Since } P_{data} \tilde{x} = \frac{1}{n} : = \sum_{i=1}^n \frac{1}{n} \log P_{\theta}(x^i)$$

$$= \frac{1}{n} \sum_{i=1}^n \log P_{\theta}(x^i)$$

$$D_{KL}(P_{data}, P_{\theta}) = \text{Entropy}(data) - \frac{1}{n} \sum_{i=1}^n \log P_{\theta}(x^i)$$

To minimize D_{KL} with respect to θ :

$$\arg \min_{\theta} D_{KL}(P_{data}, P_{\theta}) = \arg \min_{\theta} \left[\text{Entropy}(P_{data}) - \frac{1}{n} \sum_{i=1}^n \log P_{\theta}(x^i) \right]$$

Entropy(P_{data}) is constant and doesn't depend on θ

$$= \arg \min_{\theta} \left[\underbrace{\text{Entropy}(P_{data})}_{\Delta} - \frac{1}{n} \sum_{i=1}^n \log P_{\theta}(x^i) \right]$$

minimize negative is to maximize

$$= \arg \max_{\theta} \left[\underbrace{\frac{1}{n} \sum_{i=1}^n \log P_{\theta}(x^i)}_{\text{constant}} \right]$$

$$= \arg \max_{\theta} \left[\sum_{i=1}^n \log P_{\theta}(x^i) \right]$$

$$\therefore \arg \min_{\theta} D_{KL}(P_{data}, P_{\theta}) = \arg \max_{\theta} \sum_{i=1}^n \log(P_{\theta}(x^i))$$

